

SUMMARY

Full-stack AI engineer building production fintech and AI products. Built and shipped 0-to-1 applications across React/TypeScript, Python services, RAG and conversational AI, APIs, and cloud infrastructure. Led cross-functional engineering delivery and owned systems from architecture, with a focus on reliability, performance, and maintainable design.

EXPERIENCE

iMogul AI

March 2026 – May 2026

Software Engineer - Machine Learning, Rockville, USA (Remote)

- Architected and shipped the Narrative Resonance Engine (NRE), a RAG based screenplay application, by developing a hybrid retrieval pipeline that combined keyword matching with transformer based semantic and emotional vector embeddings.
- Increased Precision@3 from 42% to 68% on a labeled query set spanning 1,000+ screenplays.
- Implemented FAISS based nearest neighbor retrieval to generate Top-K candidates for thematic queries.
- Built an offline retrieval evaluation pipeline to calculate Precision@3 on labeled query screenplay pairs and compare semantic embeddings, emotional embeddings and hybrid raking configurations.

eBayXUSC

May 2025 – Aug 2025

Research Assistant – Graph Databases & Storage Systems, Los Angeles, USA

- Architected and deployed a distributed graph database on CloudLab by integrating JanusGraph with FoundationDB and Berkeley DB across 2-tier client-storage and 3-tier client-coordinator-storage configurations.
- Increased social graph query throughput by 35% and reduced average latency by 15 ms, as measured using the BG benchmark, by optimizing Gremlin traversal algorithms across multi-tier deployments.
- Automated CloudLab node provisioning, configuration, and SSH-based inter-node communication, reducing manual setup errors

HighRadius

Dec 2022 – Jun 2024

Associate Consultant 2 - Enterprise FinTech & Order-to-Cash Systems, Hyderabad, India

- Cut end-to-end processing time from 3min to 15sec by optimizing SQL queries and workflow, where I also received the Spot Award.
- Increased application throughput by 42% by parallelizing independent Order-to-Cash(O2C) workloads through multi-thread processing.
- Reduced data query latency by 30% by building an AWS-based pipeline using EC2, RDS, and S3 and optimizing database indexes.
- Developed and released 50+ enterprise software modules using object-oriented design(OOD), delivering tested production changes through CI/CD pipelines in collaboration with DevOps and QA.

PROJECTS

Echoes, Winner at LA Tech Week 2025 | Skills: Python, Flask, React.js, MediaPipe, HuBERT, REST APIs, Multimodal AI

- Engineered a real-time lip-reading pipeline using MediaPipe for lip detection, HuBERT-433h fine-tuned on a custom 20-word dataset, and cosine-similarity-based matching for therapy-specific speech recognition.
- Integrated DeepFace-based emotion detection to capture patients' affective states and correlate them with speech accuracy.
- Designed a Flask backend to perform VSR inference and PDF report generation for word accuracy, emotion, and session analysis.
- Built an interactive React and Tailwind frontend with live feedback overlays, real-time progress visualization, and automated delivery of reports to doctors and patients through an SMTP microservice.

X-GastroAI, Explainable Gastric Cancer Detector | Skills: PyTorch, TensorFlow, ResNet-50, Grad-CAM

- Built an explainable deep learning application for gastric histopathology classification by developing an image preprocessing pipeline and fine-tuning a ResNet-50 encoder through progressive unfreezing, achieving a 97.6% F1-score.
- Conducted precision-recall benchmarking and error analysis to improve model generalization, reducing false negatives by 34%, and integrated Grad-CAM feature attribution to improve model transparency.

Zariya, Winner at AthenaHacks 2026 | Skills: ASR, Gemini, ElevenLabs TTS, Voice AI

- Engineered a real-time STS (speech-to-speech) AI pipeline integrating ASR, Google Gemini, and ElevenLabs TTS (text-to-speech) through asynchronous API orchestration, reducing end-to-end response latency to under 1.5 seconds.
- Reduced end-to-end response latency to under 1.5 seconds by engineering a real-time speech-to-speech AI pipeline integrating ASR, Gemini, and ElevenLabs TTS through asynchronous API orchestration.

EDUCATION

University of Southern California, Master of Science in Computer Science | GPA: 3.7/4.0

Aug 2024 – May 2026

Courses: Advanced Data Stores, Algorithms, Security Systems, WebAssembly (WASM), AI/ML, Neural Networks

SRM University, Bachelor of Technology in Computer Science and Engineering | GPA: 3.8/4.0

Jun 2018 – May 2022

TECHNICAL SKILLS

- Programming:** Python, Java, C/C++, SQL, JavaScript/TypeScript, Shell Scripting
- ML & AI:** Deep Learning, Computer Vision, Transfer Learning, LangGraph, Prompt engineering, LLM/RAG
- Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face, OpenCV, Django, Flask, Spring Boot, React, Next.js
- Data & Storage:** PostgreSQL, MongoDB, Spark, FAISS, Qdrant
- Cloud Platforms:** AWS (S3, SageMaker, EC2, Lambda, DynamoDB), Google Cloud
- Developer Tools:** Git, GitHub, Linux, CI/CD, Unit Testing, Integration Testing, Jira, Postman, Cursor, Codex, Claude Code

PUBLICATION

- Contributing to the development of the Narrative Resonance Engine (NRE) at iMogul AI, a provisional patent-pending for semantic retrieval and narrative intelligence.